

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

COMPLEX NUMBERS

DIBY (DO IT BY YOURSELF)

By: Sachin Jakhar (Defender Mamu)





DIBY-01

Find the value of $\sum_{k=1}^{10} (i^k + i^{2k} + i^{3k} + i^{4k})$

[Ans. :]

DIBY-02

Find the value of $\left(\frac{1+i}{1-i}\right)^{10001} + \left(\frac{1-i}{1+i}\right)^{10001}$.

[Ans. : Zero]

DIBY-03

If $\frac{x-3}{3+i} + \frac{y-3}{3-i} = i$, where $x, y \in R$, then find (x, y) .

[Ans. : $y = 8, x = -2$]

DIBY-04

Find the reciprocal of $3 + \sqrt{7}i$

[Ans. : $\frac{3}{16} - \frac{\sqrt{7}}{16}i$]

DIBY-05

Find the square root of $3 - 4i$

- (A) $\pm(2 - i)$ (B) $\pm(2 + i)$ (C) $\pm(\sqrt{3} - 2i)$ (D) $\pm(\sqrt{3} + 2i)$

[Ans. : (A)]

DIBY-06

If $x + iy = \begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix}$, find x and y .

- (A) $x = 3, y = 1$ (B) $x = 1, y = 1$
(C) $x = 0, y = 3$ (D) $x = 0, y = 0$

[IIT-JEE 1998]

[Ans. : (D)]

DIBY-07

The smallest positive integer n for which $\left(\frac{1+i}{1-i}\right)^n = 1$, is

[IIT-JEE 1980]

[Ans. : 4]



DIBY-08

The conjugate of a complex number is $\frac{1}{i-1}$, then that complex number is **[AIEEE-2008]**

- (A) $\frac{-1}{i-1}$ (B) $\frac{1}{i+1}$ (C) $\frac{-1}{i+1}$ (D) $\frac{1}{i-1}$

[Ans. : (C)]

DIBY-09

If $(x - iy)^{1/3} = a + ib$, then is $\frac{x}{a} + \frac{y}{b}$ equal to **[IIT-JEE-1982]**

- (A) $4(a^2 + b^2)$ (B) $4(a^2 - b^2)$ (C) $4(b^2 - a^2)$ (D) None of these

[Ans. : (B)]

DIBY-10

If z_1, z_2, z_3 are 3 distinct complex numbers such that $\frac{3}{|z_2 - z_3|} = \frac{4}{|z_3 - z_1|} = \frac{5}{|z_1 - z_2|}$, then the value of $\frac{9}{z_2 - z_3} + \frac{16}{z_3 - z_1} + \frac{25}{z_1 - z_2}$ equals

- (A) 0 (B) 3 (C) 4 (D) 5

[Ans. : (A)]

DIBY-11

If the real part of the complex number $z = \frac{3+2i\cos\theta}{1-3i\cos\theta}$, $\theta \in \left(0, \frac{\pi}{2}\right)$ is zero, then the value of $\sin^2 3\theta + \cos^2 \theta$ is equal to

[JEE Mains-2021]

[Ans. : 01]

DIBY-12

The sum of all possible values of $\theta \in [-\pi, 2\pi]$, for which is purely imaginary $\frac{1+i\cos\theta}{1-2i\cos\theta}$, is equal to

[JEE Mains-2024]

- (A) 4π (B) 3π (C) 2π (D) 5π

[Ans. : (B)]

DIBY-13

Let $\left(-2 - \frac{1}{3}i\right)^3 = \frac{x+iy}{27}$, ($i = \sqrt{-1}$) where x and y are real numbers, then $y - x$ equals

[JEE Mains-2019]

- (A) -85 (B) -91 (C) 85 (D) 91

[Ans. : (D)]

DIBY-14

Find the real values of x and y for which the following equation is satisfied

$$\frac{(1+i)x-2i}{3+i} + \frac{(2-3i)y+i}{3-i} = i.$$

[IIT-JEE-1980]

[Ans. : $x = 3, y = -1$]

DIBY-15

Let $S = \left\{ z = x + iy : \frac{2z-3i}{4z+2i} \text{ is a real number} \right\}$. Then which of the following is NOT correct?

[JEE Mains-2023]

(A) $y + x^2 + y^2 \neq -\frac{1}{4}$

(B) $(x, y) = \left(0, -\frac{1}{2}\right)$

(C) $x = 0$

(D) $y \in \left(-\infty, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, \infty\right)$

[Ans. : (B)]

DIBY-16

The complex number $\left(\frac{1+2i}{1-i}\right)$ lies in the

[IIT-JEE-1992]

(A) Ist quadrant (B) IInd quadrant (C) IIIrd quadrant (D) IVth quadrant

[Ans. : (B)]

DIBY-17

If $z = x + iy$, $xy \neq 0$, satisfies the equation $z^2 + i\bar{z} = 0$, then $|z^2|$ is equal to:

[JEE Mains-2024]

- (A) 9 (B) $1/4$ (C) 4 (D) 1

[Ans. : (D)]

DIBY-18

The complex number z satisfies $z + |z| = 2 + 8i$. The value of $|z|$ is

- (A) 10 (B) 13 (C) 17 (D) 23

[Ans. : (C)]

DIBY-19

If $|z| = 1$ and $w = \left(\frac{z-1}{z+1}\right)$ (where $z \neq -1$), then $\text{Re}(w)$ is

[IIT-JEE-2003]

- (A) 0 (B) $\frac{1}{|z+1|^2}$ (C) $\left|\frac{1}{z+1}\right| \cdot \frac{1}{|z+1|^2}$ (D) $\frac{\sqrt{2}}{|z+1|^2}$

[Ans. : (A)]

DIBY-20

If $a^2 + b^2 = 1$, then $\left(\frac{1+b+ia}{1+b-ia}\right) =$

- (A) 1 (B) 2 (C) $b + ia$ (D) $a + ib$

[Ans. : (C)]

DIBY-21

The complex numbers $\sin x + i\cos 2x$ and $\cos x - i\sin 2x$ are conjugate to each other, for

[IIT-JEE-188]

- (A) $x = n\pi$ (B) $x = 0$ (C) $x = (n + 1/2)\pi$ (D) no value of x

[Ans. : (D)]

DIBY-22

Prove the identity, $|1 + z_1\bar{z}_2|^2 + |z_1 + z_2|^2 = (1 + |z_1|^2)(1 + |z_2|^2)$

DIBY-23

If $z^2 + 2(1 + 2i)z - (11 + 2i) = 0$, find z in the form of $a + ib$.

[Ans. : $z = 2 - i$ or $z = -(4 + 3i)$]



DIBY-24

If z and ω are two non-zero complex numbers such that $|z\omega| = 1$ and $\arg(z) - \arg(\omega) = \frac{\pi}{2}$, then $\bar{z}\omega$ is equal to [AIEEE-2003]

- (A) $-i$ (B) 1 (C) -1 (D) i

[Ans. : (A)]

DIBY-25

z and w are two nonzero complex numbers such that $|z| = |w|$ and $\arg z + \arg w = \pi$, then equals [AIEEE-2002]

- (A) $\bar{\omega}$ (B) $-\bar{\omega}$ (C) ω (D) $-\omega$

[Ans. : (B)]

DIBY-26

Let z and w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and $\arg zw = \pi$. Then $\arg z$ equals [AIEEE-2004]

- (A) $\frac{5\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{4}$

[Ans. : (C)]

DIBY-27

If $z = \frac{1}{2} - 2i$ is such that $|z + 1| = \alpha z + \beta(1 + i)$, $i = \sqrt{-1}$ and $\alpha, \beta \in R$, then $\alpha + \beta$ is equal to

[JEE Mains-2024]

- (A) -4 (B) 3 (C) 2 (D) -1

[Ans.: (B)]

DIBY-28

Let z_1 and z_2 be two complex number such that $z_1 + z_2 = 5$ and $z_1^3 + z_2^3 = 20 + 15i$. Then $|z_1^4 + z_2^4|$ equals

[JEE Mains-2024]

- (A) $30\sqrt{3}$ (B) 75 (C) $15\sqrt{15}$ (D) $25\sqrt{3}$

[Ans.: (B)]

DIBY-29

Let $A = \{z \in C: |z - 2 - i| = 3\}$, $B = \{z \in C: \operatorname{Re}(z - iz) = 2\}$ and $S = A \cap B$. Then $\sum_{z \in S} |z|^2$ is equal to

[JEE Mains-2025]

[Ans.: 22]



DIBY-30

Solve $z^2 + |z| = 0$

[Ans. : $(0, 0)(0, 1)(0, -1)$]

DIBY-31

If $|z - i| = 1$ and $\text{Arg}(z) = \frac{\pi}{2}$, then find z .

[Ans. : $2i$]

DIBY-32

Write the complex number $\frac{(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3})(\sqrt{3} + i)}{i - 1}$ in

[Ans. : $\sqrt{2}\text{cis}\left(\frac{-11\pi}{12}\right)$]

DIBY-33

If $z = \frac{1 + i\sqrt{3}}{2i(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})}$, find $|z|$ and $\text{Arg}(z)$.

[Ans. : 1 & $-\frac{\pi}{2}$]

DIBY-34

Simplify $\left(\frac{1 + i \sin \frac{\pi}{8} + \cos \frac{\pi}{8}}{1 - i \sin \frac{\pi}{8} + \cos \frac{\pi}{8}} \right)^8$

[Ans. : -1]

DIBY-35

The value of $\left(\frac{-1 + i\sqrt{3}}{1 - i} \right)^{30}$ is

- (A) $-2^{15}i$ (B) -2^{15} (C) $2^{15}i$ (D) 65

[JEE (Main)-2020]

[Ans. :]

DIBY-36

If a and b are real numbers such that $(2 + \alpha)^4 = a + b\alpha$, where $\alpha = \frac{-1 + i\sqrt{3}}{2}$, then $a + b$ is equal to

[JEE (Main)-2020]

- (A) 33 (B) 9 (C) 24 (D) 57

[Ans. :]

DIBY-37

Let $\left| \frac{\bar{z}-i}{2\bar{z}+i} \right| = \frac{1}{3}, z \in \mathbb{C}$, be the equation of a circle with center at C . If the area of the triangle, whose vertices are at the points $(0, 0)$, C and $(\alpha, 0)$ is 11 square units, then α^2 equals **[JEE Mains-2025]**

- (A) 100 (B) $\frac{81}{25}$ (C) $\frac{121}{25}$ (D) 50

[Ans. : (A)]

DIBY-38

Let a, b, x and y be real numbers such that $a - b = 1$ and $y \neq 0$. If the complex number

$z = x + iy$ satisfies $\text{Im} \left(\frac{az+b}{z+1} \right) = y$, then which of the following is(are) possible value(s) of x ? **[JEE (Adv.)-2017, Paper-1]**

- (A) $-1 + \sqrt{1 - y^2}$ (B) $1 - \sqrt{1 + y^2}$ (C) $1 + \sqrt{1 + y^2}$ (D) $-1 - \sqrt{1 - y^2}$

[Ans. : (A), (D)]

DIBY-39

Let $z = x + iy$ be a complex number where x and y are integers. Then the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$ is

[IIT-JEE-2009, Paper-1]

- (A) 48 (B) 32 (C) 40 (D) 80

[Ans. : (A)]

DIBY-40

Let z be a complex number such that the imaginary part of z is non-zero and $a = z^2 + z + 1$ is real. Then a cannot take the value

[IIT-JEE-2012]

- (A) -1 (B) 1/3 (C) 1/2 (D) 3/4

[Ans. : (D)]

DIBY-41

Let $A = \left\{ \frac{1967 + 1686i \sin \theta}{7 - 3i \cos \theta} : \theta \in \mathbb{R} \right\}$. If A contains exactly one positive integer n , then the value of n is

[JEE (Adv.)-2023]

[Ans. : 281]



DIBY-42

If the value of $\sum_{p=1}^{32} (3p + 2) \left\{ \sum_{q=1}^{10} \left(\sin \frac{2q\pi}{11} - i \cos \frac{2q\pi}{11} \right) \right\}^p$ is $a + ib$, then find the value of $a + b$ _____.

[Ans. : 0]

DIBY-43

For a non-zero complex number z , let $\arg(z)$ denote the principal argument of z , with $-\pi < \arg(z) \leq \pi$. Let ω be the cube root of unity for which $0 < \arg(\omega) < \pi$. Let $\alpha = \arg\left(\sum_{n=1}^{2025} (-\omega)^n\right)$. Then the value of $\frac{3\alpha}{\pi}$ is.

[JEE (Adv.)-2025]

[Ans. : -2]

DIBY-44

If α is the n^{th} root of unity, then $1 + 2\alpha + 3\alpha^2 + 4\alpha^3 + \dots$ upto n terms is equal to:

- (A) $\frac{-n}{(1-\alpha)^2}$ (B) $\frac{-n}{1-\alpha}$ (C) $\frac{-2n}{1-\alpha}$ (D) $\frac{-2n}{(1-\alpha)^2}$

[Ans. : (B)]

DIBY-45

Let $i = \sqrt{-1}$. If $\frac{(-1+i\sqrt{3})^{21}}{(1-i)^{24}} + \frac{(1+i\sqrt{3})^{21}}{(1+i)^{24}} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$.

Then $\sum_{j=0}^{n+5} (j+5)^2 - \sum_{j=0}^{n+5} (j+5)$ is equal to _____.

[JEE Mains-2021]

[Ans. : 310]

DIBY-46

If $\alpha, \beta \in R$ are such that $1 - 2i$ (here $i^2 = -1$) is a root of $z^2 + \alpha z + \beta = 0$, then $(\alpha - \beta)$ is equal to

[JEE Mains-2021]

[Ans. : -7]

DIBY-47

Let z_0 be a root of the quadratic equation, $x^2 + x + 1 = 0$. If $z = 3 + 6iz_0^{81} - 3iz_0^{93}$, then $\arg z$ is equal to

[JEE Mains-2019]

(A) $\pi/4$

(B) $\pi/3$

(C) 0

(D) $\pi/6$

[Ans. : (A)]

DIBY-48

If a and b are real numbers such that $(2 + \alpha)^4 = a + b$, where $\alpha = \frac{-1+i\sqrt{3}}{2}$, then $a + b$ is equal to

[JEE Mains-2020]

- (A) 33 (B) 9 (C) 24 (D) 57

[Ans. :]

DIBY-49

The number of common roots (complex or real both) in the equation

$$z^{12} - 1 = 0, z^{18} - 1 = 0 \text{ and } z^{24} - 1 = 0 \text{ is}$$

[Ans. : 6]

DIBY-50

Let $|z_1 - 8 - 2i| \leq 1$ and $|z_2 - 2 + 6i| \leq 2, z_1, z_2 \in \mathbb{C}$. Then the minimum value of $|z_1 - z_2|$ is:

[JEE Mains-2025]

- (A) 13 (B) 3 (C) 7 (D) 10

[Ans. : (c)]



DIBY-51

The number of elements in the set $\{z = a + ib \in \mathbb{C} : a, b \in \mathbb{Z} \text{ and } 1 < |z - 3 + 2i| < 4\}$ is

[JEE Mains-2022]

[Ans. : 40]

DIBY-52

If the four complex numbers and $z, \bar{z}, \bar{z} - 2\text{Re}(\bar{z}), z - 2\text{Re}(z)$ represent the vertices of a square of side 4 units in the Argand plane, then $|z|$ is equal to:

[JEE Mains-2020]

(A) $4\sqrt{2}$

(B) 2

(C) $2\sqrt{2}$

(D) 4

[Ans. : (C)]

DIBY-53

Let z be a complex number such that $|z| = 1$. If $\frac{2+k^2z}{k+\bar{z}} = kz, k \in \mathbb{R}$, then the maximum distance of $k + ik^2$ from the circle $|z - (1 + 2i)| = 1$ is: [JEE Mains-2025]

- (A) $\sqrt{5} + 1$ (B) 2 (C) 3 (D) $\sqrt{3} + 1$

[Ans.: (A)]

DIBY-54

If $S = \{z \in \mathbb{C} : |z - i| = |z + i| = |z - 1|\}$, then, $n(S)$ is: [JEE Mains-2024]

- (A) 1 (B) 0 (C) 3 (D) 2

[Ans.: (A)]

DIBY-55

Let z be a complex number such that $\left| \frac{z-2i}{z+i} \right| = 2, z \neq -i$. Then z lies on the circle of radius 2 and centre [JEE Mains-2023]

- (A) $(2, 0)$ (B) $(0, 0)$ (C) $(0, 2)$ (D) $(0, -2)$

[Ans.: (D)]

DIBY-56

Locus of complex number satisfying $\arg[(z - 5 + 4i)/(z + 3 - 2i)] = -\pi/4$ is the arc of a circle

(A) whose radius is $5\sqrt{2}$

(B) whose radius is 5

(C) whose length (of arc) is $\frac{15\pi}{\sqrt{2}}$

(D) whose centre is $-2 - 5i$

[Ans. : (A), (C), (D)]

Match the following

Column-I

Column-II

A	$ z - 6i + z - 8 = k$ will represent an ellipse for k equals to	(p)	2
B	$ z - 12i + 3 - z - 2 = k$ will represent hyperbola if k equals to	(q)	8
C	$ z - ki + z - 4 = \sqrt{10k}$ will represent line segment if k equals to	(r)	12
D	$\frac{ z - k + 2ki }{ z - 2 + 4i } = k$ will represent circle if k equals to	(s)	11
		(t)	10

[Ans. : $A \rightarrow r, s$; $B \rightarrow p, q, r, s, t$; $C \rightarrow p, q$; $D \rightarrow q, r, s, t$]



DIBY-58

Let z be a complex number such that $|z - i| - |z + 5i| = 0$ and $|z| = 2$. Then

[JEE Mains-2022]

(A) $x + 2y = 4$

(B) $x^2 + y = 4$

(C) $x + 2y = -4$

(D) $x^2 - y - 3 = 0$

[Ans. : (c)]

DIBY-59

If complex numbers z_1, z_2, z_3 respectively represent the vertices of an isosceles right angled triangle ABC with right angle at B then show that: $z_1^2 + 2z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$

DIBY-60

Let z_i (where $i = 1, 2, 3, 4, 5, 6$) represents the vertices of a regular hexagon in the complex plane and z_0 is its centre. If $\sum_{i=1}^6 z_i^2 = k \cdot z_0^2$, find the value of k .

[Ans. : 6]

DIBY-61

If $S_1 = \{z \in \mathbb{C} : 3 \leq |z - 2 - 2i| \leq 6\}$, $S_2 = \left\{z \in \mathbb{C} : \frac{\pi}{6} \leq \arg(z - 2 - 2i) \leq \frac{\pi}{2}\right\}$

then find area enclosed by $S_1 \cap S_2$.

[Ans. :]

Let $a, b \in \mathbb{R}$ and $a^2 + b^2 \neq 0$. Suppose $S = \left\{ z \in \mathbb{C} : z = \frac{1}{a+ibt}, t \in \mathbb{R}, t \neq 0 \right\}$, where $i = \sqrt{-1}$. If $z = x + iy$ and $z \in S$, then (x, y) lies on

- A** The circle with radius $\frac{1}{2a}$ and centre $\left(\frac{1}{2a}, 0\right)$ for $a > 0, b \neq 0$
- B** The circle with radius $-\frac{1}{2a}$ and centre $\left(-\frac{1}{2a}, 0\right)$ for $a < 0, b \neq 0$
- C** The x-axis for $a \neq 0, b = 0$
- D** The y-axis for $a = 0, b \neq 0$

[Ans. : (A,C,D)]

Let S be the set of all complex numbers z satisfying $|z - 2 + i| \geq \sqrt{5}$.

If the complex Number z_0 is such that $\frac{1}{|z_0 - 1|}$ is the maximum of the set

$\left\{ \frac{1}{|z - 1|} : z \in S \right\}$, then the principal argument of $\frac{4 - z_0 - \bar{z}_0}{z_0 - \bar{z}_0 + 2i}$ is

- A $\frac{\pi}{2}$
- B $\frac{\pi}{4}$
- C $-\frac{\pi}{2}$
- D $\frac{3\pi}{4}$

[Ans. : (c)]



DIBY-64

Let $A_1, A_2, A_3, \dots, A_8$ be the vertices of a regular octagon that lie on a circle of radius 2. Let P be a point on the circle and let PA_i denote the distance between the points P and A_i for $i = 1, 2, \dots, 8$. If P varies over the circle, then the maximum value of the product $PA_1 \cdot PA_2 \dots PA_8$, is

[JEE (Adv.)-2023]

[Ans. : 512]

DIBY-65

Let $f(x) = \frac{\sum_{r=0}^{100} {}^{100}C_r \sin 2rx}{\sum_{r=0}^{100} {}^{100}C_r \cos 2rx}$, then $f\left(\frac{\pi}{800}\right)$ is equal to

[CHALLENGER]

- (A) $\sqrt{2} + 1$ (B) $\sqrt{2} - 2$ (C) $\sqrt{2} - 1$ (D) $2\sqrt{2} - 2$

[Ans. : (c)]

DIBY-66

[CHALLENGER]

Factorise $z^7 - 1 = 0$ into real linear and quadratic factors.

Also find value of $\cos \frac{\pi}{7} \cdot \cos \frac{2\pi}{7} \cdot \cos \frac{4\pi}{7} = ?$

[Ans. : -1/8]

DIBY-67

Let z_1, z_2 be the roots of the equation $z^2 + az + 12 = 0$ and z_1, z_2 form an equilateral triangle with origin. Then, the value of $|a|$ is

[JEE Mains-2021]

[Ans.: 6]

DIBY-68

If $z_1, z_2, z_3 \in \mathbb{C}$ are the vertices of an equilateral triangle, whose centroid is z_0 , then $\sum_{k=1}^3 (z_k - z_0)^2$ is equal to

[JEE Mains-2025]

- (A) 0 (B) 1 (C) i (D) $-i$

[Ans.: (A)]

DIBY-69

Let O be the origin and A be the point $z_1 = 1 + 2i$. If B is the point z_2 , $\text{Re}(z_1) < 0$, such that OAB is a right-angled isosceles triangle with OB as hypotenuse, then which of the following is NOT true?

[JEE Mains-2022]

- (A) $\arg z_2 = \pi - \tan^{-1} 3$ (B) $\arg(z_1 - 2z_3) = -\tan^{-1}(4/3)$
 (C) $|z_2| = \sqrt{10}$ (D) $|2z_1 - z_2| = 5$

[Ans.: (D)]

DIBY-70

Let O be the origin, the point A be $z_1 = \sqrt{3} + 2\sqrt{2}i$, the point $B(z_2)$ be such that $\sqrt{3}|z_2| = |z_1|$ and $\arg(z_2) = \arg(z_1) + \frac{\pi}{6}$. Then: **[JEE Mains-2025]**

- (A) area of triangle ABO is $\frac{11}{\sqrt{3}}$
- (B) ABO is an obtuse angled isosceles triangle
- (C) area of triangle ABO is $\frac{11}{4}$
- (D) ABO is a scalene triangle

[Ans. : (B)]

DIBY-71

If $\operatorname{Re}\left(\frac{z-1}{2z+i}\right) = 1$; where $z = x + iy$; then the point (x, y) lies on a: **[JEE Mains-2022]**

- (A) straight line whose slope is $-2/3$
- (B) straight line whose slope $3/2$
- (C) circle whose diameter is $\sqrt{5}/2$
- (D) circle whose centre is at $(-1/2, -3/2)$

[Ans. : (C)]

DIBY-72

If z be a complex number satisfying $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| = 4$, then $|z|$ cannot be

[JEE Mains-2020]

- (A) $\sqrt{10}$ (B) $\sqrt{8}$ (C) $\sqrt{7}$ (D) $\frac{\sqrt{17}}{2}$

[Ans. : (C)]

DIBY-73

Let z be a complex number such that the real part of $\frac{z-2i}{z+2i}$ is zero. Then, the maximum value of $|z - (6 + 8i)|$ is equal to

[JEE Mains-2024]

- (A) 12 (B) 10 (C) 8 (D) ∞

[Ans. : (A)]

DIBY-74

Let z be those complex number which satisfy

$$|z + 5| \leq 4 \text{ and } z(1 + i) + \bar{z}(1 - i) \geq -10, i = \sqrt{-1}.$$

If the maximum value of $|z + 1|^2$ is $\alpha + \beta\sqrt{2}$, then the value of $(\alpha + \beta)$ is ____.

[JEE mains-2021]

[Ans. : 48]

DIBY-75

Let $S_1 = \{z \in \mathbb{C} : |z| \leq 5\}$, $S_2 = \left\{z \in \mathbb{C} : \operatorname{Im} \left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i} \right) \geq 0 \right\}$ and $S_3 = \{z \in \mathbb{C} : \operatorname{Re}(z) \geq 0\}$. Then the area of the region $S_1 \cap S_2 \cap S_3$ is: [JEE Mains-2024]

- (A) $\frac{125\pi}{12}$ (B) $\frac{125\pi}{4}$ (C) $\frac{125\pi}{24}$ (D) $\frac{125\pi}{6}$

[Ans. : (A)]

DIBY-76

Let $S = \{z \in \mathbb{C} : |z - 2| \leq 1 \text{ and } z(1+i) + \bar{z}(1-i) \leq 2\}$. Let $|z + 4i|$ attains minimum and maximum values, respectively, at $z_1 \in S$ and $z_2 \in S$. If $5(|z_1|^2 + |z_2|^2) = \alpha + \beta\sqrt{5}$, where α and β are integers, then the value of $\alpha + \beta$ is equal to ____ . [JEE Mains-2022]

[Ans. : 26]

DIBY-77

Let $A = \{z \in \mathbb{C} : |z - 2 - i| = 3\}$, $B = \{z \in \mathbb{C} : \operatorname{Re}(z - iz) = 2\}$ and $S = A \cap B$. Then $\sum_{z \in S} |z|^2$ is equal to ____ . [JEE Mains-2025]

[Ans. : 22]



DIBY-78

Let $S = \{z \in \mathbb{C} : |z - 1| = 1 \text{ and } (\sqrt{2} - 1)(z + \bar{z}) - i(z - \bar{z}) = 2\sqrt{2}\}$. Let $z_1, z_2 \in S$ be such that

$|z_1| = \max_{z \in S} |z|$ and $|z_2| = \min_{z \in S} |z|$. Then $|\sqrt{2}z_1 - z_2|^2$ equals: [JEE Mains-2024]

- (A) 1 (B) 4 (C) 3 (D) 2

[Ans. : (D)]

DIBY-79

If the locus of $z \in \mathbb{C}$, such that $\operatorname{Re}\left(\frac{z-1}{2z+i}\right) + \operatorname{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right) = 2$ is a circle of radius r and center (a, b) , then $\frac{15ab}{r^2}$ is equal to: [JEE Mains-2025]

- (A) 24 (B) 12 (C) 18 (D) 16

[Ans. : (C)]

DIBY-80

If the area of the triangle in Argand diagram formed by $z, wz, z + wz$ where ' w ' ($w \neq 1$) is cube root of unity is $16\sqrt{3}$ sq. units, then $|z|$ is ____ . [Ans. : 8]

DIBY-81

Let $\mathbb{R}$ denote the set of all real numbers. Let $z_1 = 1 + 2i$ and $z_2 = 3i$ be two complex numbers, where $i = \sqrt{-1}$. Let $S = \{(x, y) \in \mathbb{R} \times \mathbb{R} : |x + iy - z_1| = 2|x + iy - z_2|\}$. Then which of the following statements is (are) TRUE? **[JEE (Adv.)-2025]**

- (A) S is a circle with centre $\left(-\frac{1}{3}, \frac{10}{3}\right)$ (B) S is a circle with centre $\left(\frac{1}{3}, \frac{8}{3}\right)$
 (C) S is a circle with radius $\frac{-\sqrt{2}}{3}$ (D) S is a circle with radius $\frac{2\sqrt{2}}{3}$

[Ans. : A, D]

DIBY-82

Let C be the circle in the complex plane with centre $z_0 = \frac{1}{2}(1 + 3i)$ and radius $r = 1$. Let $z_1 = 1 + i$ and the complex number z_2 be outside circle C such that $|z_1 - z_0||z_2 - z_0| = 1$. If z_0, z_1 and z_2 are collinear, then the smaller value of $|z_2|^2$ is equal to: **[JEE Mains-2023]**

- (A) $\frac{5}{2}$ (B) $\frac{7}{2}$ (C) $\frac{13}{2}$ (D) $\frac{3}{2}$

[Ans. : (A)]



*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**